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A closed loop 2.65 K hybrid JT cooler for future space application

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Cryogenics and its applications in space have made a remarkable amount of progresses over the last few decades. In the area of space astronomy, some electronic detectors should be cooled to 1–4 K to improve their performance [1,2]. So the miniature 1–4 K cryocooler for space application has become a key field for cryogenic research in recent years. In view of their flexibility and relatively higher efficiency, hybrid JT cryocoolers are widely used in space detectors working at below 4 K. In fact, nearly all the space applications of mechanical cryocoolers working at the temperature of 1–4 K, having been launched or under development are hybrid JT cryocoolers [2–6].

Recently, Superconducting Nanowire Single Photon Detectors (SNSPD) have demonstrated features of high quantum efficiency, high counting rate, low timing jitter, low dark count rate, and wide spectrum sensitivity, which shows promising prospects in space applications. Thus, a compact and efficient space cryocooler which is able to achieve temperatures below 3 K is required. Narasaki et al. [7] have developed a 1.7 K hybrid JT cooler consisting of a ^{3}He JT cooler and a two-stage Stirling cooler. And the performance of a ^{4}He JT cooler is also tested. Kotsubo et al. [8] supported by National Institute of Standards and Technology (NIST) have developed a cooling system for SNSPD. The hybrid cooler consists of a ^{4}He JT cooler operating at 2.2 K and a 3-stage pulse tube cooler operating at 35 Hz. But the JT compressor suitable for space application has not been developed. In China, the SNSPDs developed by Shanghai Institute of Microsystem and Information Technology (SIMIT), Chinese Academy of Sciences (CAS) have also demonstrated outstanding performance. The SNSPD system for space applications can operate sufficiently well below 3 K [9] and the heat load of the SNSPDs is lower than 1 mW. So a cooling system operating below 3 K is in urgent demand.

The Key Laboratory of Space Energy Conversion Technologies (SECT) of the Technical Institute of Physics and Chemistry (TIPC), CAS has previously developed a 4.5 K hybrid JT cryocooler [10]. Based on this cooler, a closed loop JT cooler operating below 3 K is designed, manufactured and tested.

The hybrid JT cooler is composed of two thermally coupled parts: the two-stage high frequency pulse tube (PT) cryocooler and the JT loop, as illustrated in Figure 1. The two cold fingers of the pulse tube cooler, precooling JT loop at 65 and 15 K respectively, are of thermally coupled coaxial configuration. The JT loop is composed of JT compressors, counter-flow heat exchangers, precooling heat exchangers, a JT valve and an evaporator. Both pulse tube cryocooler and JT loop are driven by oil-free linear compressors, which makes the hybrid JT cooler suitable for space applications. The JT loop is driven by a three-stage compression system. High pressure ^{4}He gas passes through the inner tube of the three counter-flow heat exchangers where it is cooled by the low pressure return gas. And two additional heat exchangers mounted on the cold end of the precooler to provide precooling power for JT loop. The JT orifice is a detachable calibrated restriction which allows the high pressure gas to

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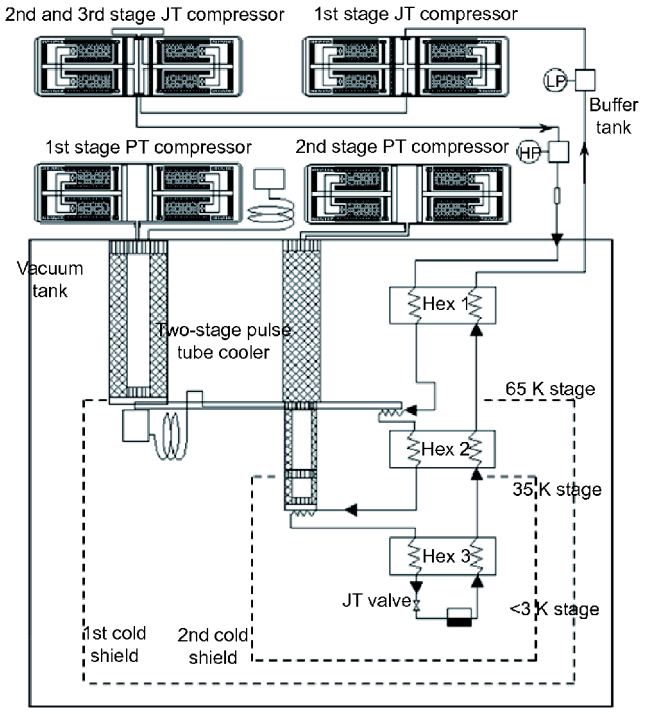


Figure 1 Schematic of the hybrid JT cooler.

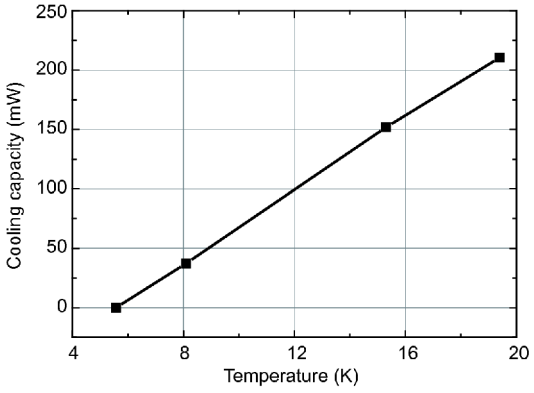


Figure 2 Performance of the pulse tube second stage cold head

Table 1 Saturation points of helium at equilibrium

Temperature (K)	Pressure (MPa)
2.17	0.00486
2.5	0.010
3.0	0.0237
4.0	0.081
4.2	0.0984

expand into low pressure JT return line. When helium gas goes through the JT valve, a fraction of the helium gas will be liquefied during the expansion process.

Besides, buffer tanks are installed at the inlet and outlet of the JT compressors to provide stable pressure ratio for JT loop. One cold shield is mounted on cold end of the first stage pulse tube cooler and another one is mounted on the mid-bypass heat exchanger of the second stage pulse tube to decrease heat load on the second stage cold end.

The high frequency pulse tube cooler is characterized by small size, long life, and low vibration and is hence widely used in space detection. The two-stage pulse cooler used in this hybrid cooler is thermally coupled configuration. The cooling capacity of the first stage pulse tube is 6 W@60 K. The performance of the second stage cold end is presented in Figure 2. Cooling power of 150 mW@15 K was achieved when charge pressure is 2.0 MPa.

The three-stage JT compressor system operates at a suction pressure much lower than atmospheric pressure, which enables cooling temperature lower than 3 K to be achieved. Table 1 gives the saturation points of helium at equilibrium. So one of the main challenges of the 2 K-class JT cooler is to achieve stable return pressure lower than 23 kPa. However, gaining such low pressure is quite difficult for space JT compressors.

For the purpose of achieving temperature below 3 K, great efforts are made to reduce the return gas pressure of the JT loop. The JT compressors play a key role in achieving lower return gas pressure. And the development of space JT compressor is the main difficulty in the development of 2 K-class JT cooler. Based on the 4.5 K JT cooler [10], a new three-stage JT compressor system was designed for the JT loop. The valve and the plate spring of the JT compressors were optimized to suit the low suction pressure. Besides, the three counter-flow heater exchangers were spiral tube-in-tube design, and channels of the return gas were optimized to reduce their pressure drop. The JT valve consists of two detachable 30 μm orifices which were arranged in tandem in the detachable flange to increase the resistance of the JT valve, as illustrated in Figure 3. If it is relatively easier for the JT compressors to gain lower return gas pressure, lower temperature can be achieved. Besides, relative larger size orifices can be used compared with single orifice which can prevent clogging up due to contamination.

Then, the optimized JT cooler precooled by the two-stage pulse tube cooler was experimentally studied. The mass flow rate of the JT cooler is 1.2 mg/s which is lower than the design value of 1.5 mg/s. Although the mass flow rate is affected by many factors, this can be mainly attributed to both the pressure of the returning gas and the fact that the inlet temperature of the JT valve is higher than the design value. And the efficiency of the three heat exchangers is larger than the design value of 97%. The temperature of the inlet and outlet gas of the heat exchangers was measured and the efficiency of the three heat exchangers (97.2%, 97.3% and 97.2% respectively) is calculated based on the measured temperature. The pressure at each stage was measured, as shown in Figure 4. The suction pressure of the first stage JT

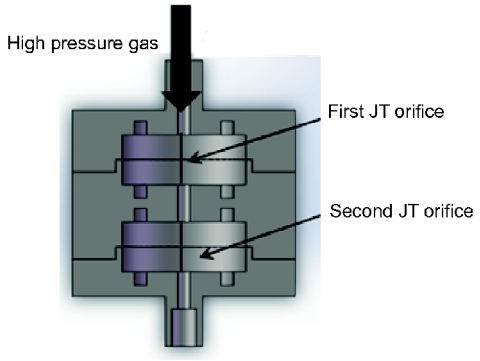


Figure 3 (Color online) Schematic of the two-stage throttling.

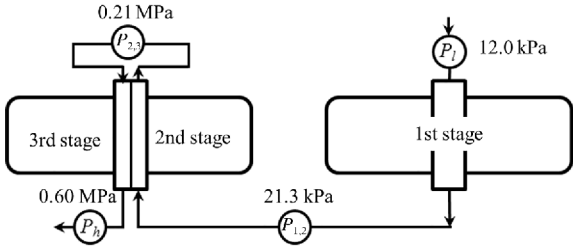


Figure 4 Schematic of three-stage compression.

compressor is higher than design value (9 kPa) which leads to the fact that the base temperature is a little higher than design value of 2.5 K. The helium in the evaporator is assumed to be in saturated state. So based on the temperature measured at the evaporator (2.65 K) and the suction pressure (12.0 kPa) of the first stage JT compressor, the pressure drop in the low pressure side is about 1.3 kPa.

Besides, we find that the pressure ratio of the three-stage compressors differs from each other and the first stage pressure ratio is quite low. Thus, specific research should be conducted on characteristics of the JT compressor operating at low pressure as well as the multi-stage compression system.

Detailed operating parameters of one of the tests are presented in Table 2. The two-stage pulse tube cooler, operating at 42 and 18 Hz respectively, precooled JT loop at 64.5 and 14.6 K. The total power consumption of the hybrid JT cooler was 317.3 W (278.6 W for PT cooler and 38.7 W for JT cooler). Temperature of 2.65 K was achieved when the high side and low side pressure were 0.6 MPa and 12 kPa respectively. And the no load temperature at the JT evaporator was quite stable, as illustrated in Figure 5, the temperature fluctuation was lower than ±0.05 K. Besides, cooling capacity of 1.48 mW was measured at 2.71 K. The measured cooling capacity is lower than design value of 4 mW at 2.7 K which can be mainly attributed to the lower mass flow rate, higher suction pressure and the heat leakage. And the total weight of the cryocooler system is about 55 kg.

In fact, series of pulse tube cryocoolers designed and manufactured by our laboratory are operating stably in the space satellites. And the JT compressors use the technology of pulse tube compressors which have been successfully used in space satellites [11]. In fact, this hybrid JT cooler is the first 2 K-class closed loop miniature ^{4}He mechanical cryocooler suitable for space application. Therefore, this hybrid JT cooler paves a way for the future space applications of SNSPDs and other detectors requiring 2 K-class temperature. However, there is still room for improvement considering the relatively large size, heavy weight and low efficiency of the present cryocooler. Future work includes further optimization of the JT compressors and the development of two-stage gas-couple pulse tube cooler. And a

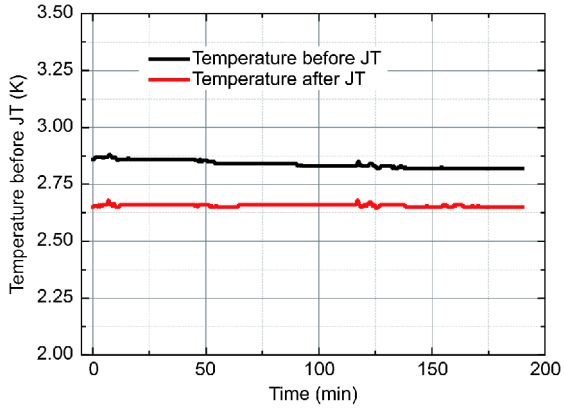


Figure 5 (Color online) The temperature fluctuation at evaporator of the JT cooler.

Table 2 Performance of the hybrid JT cooler

Operating parameters of the hybrid cooler		First stage	Second stage	Third stage
Pulse tube cooler	Input power (W)	170.2	108.4	–
	Frequency (Hz)	42	18	–
	Precooling temperature (K)	64.5	14.6	–
JT cooler	Input power (W)	8.4	9.8	20.5
	Frequency (Hz)	50	50	50
	Pressure ratio P_h/P_l (MPa/MPa)		0.6/0.0.012	
Base temperature of the JT cooler			2.65 K	

more compact and efficient 2.0 K hybrid JT cooler is under development.

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1 Ross Jr R G, Boyle R F, Kittel P. NASA space cryocooler programs—a 2003 overview. In: Proceedings of the International Thermal Detectors Workshop. 2003, 710: 1197–1204
2 Coulter D R, Ross Jr R G, Boyle R F, et al. NASA advanced cryocooler technology development program. Astronomical Telescopes and Instrumentation. In: Proceedings of the International Society for Optics and Photonics. SPIE, 2003. 1020–1028
3 Ross Jr R G. A Study of the Use of 6K ACTDP Cryocoolers for the MIRI Instrument on JWST. Boston: Springer, 2005
4 Otsuka K, Tsunematsu S, Okabayashi A, et al. Test results after refurbish of cryogenic system for smiles. Cryogenics, 2010, 50: 512–515
5 Lee I Y S, Suzuki H. Energy transfer and amplified spontaneous emission in temperature-controlled random scattering media. J Phys Chem B, 2008, 112: 4561–4570
6 Sugita H, Sato Y, Nakagawa T, et al. Cryogenic system design of the next generation infrared space telescope SPICA. Cryogenics, 2010, 50: 566–571
7 Narasaki K, Tsunematsu S, Kanao K, et al. Mechanical coolers operating below 4.5 K for space application. Nucl Instruments Methods Phys Res Sect A-Accelerators Spectrometers Detectors Associated Equipment, 2006, 559: 644–647
8 Kotsubo V, Radebaugh R, Hendershott P, et al. Compact 2.2 K cooling system for superconducting nanowire single photon detectors. IEEE Trans Appl Supercond, 2017, 27: 1–5
9 You L, Quan J, Wang Y, et al. Superconducting nanowire single photon detection system for space applications. Opt Express, 2018, 26: 2965–2971
10 Quan J, Zhou Z J, Liu Y J, et al. A miniature liquid helium temperature JT cryocooler for space application. Sci China Technol Sci, 2014, 57: 2236–2240
11 Cai J H, Zhao M G, Hong G T. The pulse tube cryocooler of GF-4 satellite staring camera (in Chinese). Spacecraft Recovery Remote Sens, 2016, 37: 66–71